

Caleb Bellsmith

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TECHNICAL SKILLS

Languages: C++, C, Python, HTML, LaTeX, Visual Basic, MATLAB, CSS, JavaScript

Tools: SolidWorks, Fusion360, AutoCAD, VSCode, KiCad, CLion, Bamboo, Arduino IDE, GitHub, GitLab

Fabrication: CNC Machining, 3D Printing, Soldering, Manual Prototyping, Woodworking, Welding, Waterjetting, Lazer Cutting

EXPERIENCE

Alchemy Nano

Lab Automation Engineer Co-op

April 2026 – September 2026

- Designed and manufactured custom automated testing equipment to enable higher precision and new styles of testing for nanocomposite film samples through custom full stack designs.
- Manufactured autonomous robotics to increase lab testing efficiency by up to 450% and lab accuracy by 40%.

Part Time Automation Engineer

Sept 2026 – Present

- Optimized and improved previously built automation systems for production grade use.
- Designed, prototyped, and manufactured a custom in line scanning system for nano-film quality assurance.

UWaterloo Space Soldering Team | Mechanical Designer

2025 – Present

- Constructing a centrifugal structure for a microgravity soldering experiment selected for the CAN-RGX Flight Campaign 2026 (NRC & CSA partnership). Technology is patent pending with commercial applications in space manufacturing and repair.
- Personally, designed and fabricated the rotation mechanism, wire feeding assembly, and general motor assembly for the centrifuge, while running preliminary prototyping tests on the design.

Undergraduate Research Assistant | Wireless Sensors and Devices Laboratory, UWaterloo

September 2026 - Present

- Designed and improved autonomous assembly line robotics learning systems, including weight and ergonomic optimization as well as PCB design.

True North Bio-competition | Electrical Designer

2025 – 2026

- Designed a custom PCB which served as the complete electrical control framework for a lower limb exoskeleton. The PCB utilized EMG and IMU sensors to map and interpret user movement into athletic recovery data.

PROJECTS

EMG Controlled Wireless Mouse

IMU & EMG Sensors, Web Dev

- Created a wearable human machine interface using an ESP32 and IMU to translate arm tilt into mouse movement and EMG sensors to trigger clicks via muscle flexion.
- Developed a full stack web dashboard to monitor device state and calibrate user specific parameters including sensitivity, Bluetooth status, and muscle group mapping.

Precision Robotic Arm

ESP32, C++, Servo Control

- Designed and hand built a custom 5 DoF robotic arm, including circuit design and assembly, high pressure grip systems, and fully 3D printed bearings.
- Currently being integrated with the EMG wireless arm sleeve to enable motion mirroring, translating human arm movement directly into real time robotic actuation.

Custom Cedar Strip Canoe

CAD, CNC Machining

- Computer modeled and fabricated a custom canoe; integrated CNC machined internal molds with traditional woodworking to achieve a smooth seamless finish.

Auto Solving Maze

ESP32, C++, Computer Vision

- Developed a servo powered tilting platform and Integrated a webcam based computer vision system to enable autonomous maze solving and real time path finding, as well as an ML model to optimize solve time.

EDUCATION & LEADERSHIP

University of Waterloo | Candidate for B.A.Sc. in Biomedical Engineering Co-op

Sept 2025 – Present

- **WEFF Representative:** Nominated by class to evaluated technical proposals for student project funding and represented class interests.

Lifeguard & Swimming Instructor | Instructor, NLS, CPR Certified

2023 – 2025

- Selected for competitive outdoor pool placement (Top 4% of 250+ staff) based on performance and leadership.

Distinctions and Awards

- Governor General's Academic Medal, Sir Isaac Newton Physics Award, TDCH Valedictorian, HHCS Valedictorian.